Accuracy and Imbalance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Accuracy and Imbalance before opening notes. Show any later correction separately.

Try the **recall** before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Train a Classifier

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Baselines First

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Accuracy and Imbalance trace and record the first step you can explain without help.

Use W2: Compare both class fractions for **Accuracy** and **class imbalance**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Accuracy and Imbalance explanation against the supplied facts before discussing it.

Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Validation has90 healthy and10 spotted cases. Always-healthy predicts all healthy. Another model has TP7,FN3,FP8,TN82. Compare **accuracy** and spotted **recall**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Accuracy and Imbalance target to a trace and a justified conclusion.

Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Accuracy and Imbalance case and show the deciding evidence.

A dataset has 95 healthy and 5 spotted. Model predicts healthy for all.

Calculate and critique **accuracy**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A classifier predicts healthy for all 100 plants; 95 are healthy and 5 diseased. Compute **accuracy** and diseased **recall**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 16 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Accuracy and Imbalance adds to Responsible Plant Classifier and keep the evidence.

Check Plant Classifier class balance and select metrics that keep the spotted-leaf class visible.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Score the majority baseline

> Score the majority baseline. Apply the supplied Accuracy and Imbalance facts and keep your reasoning.

Training has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, **accuracy**, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Compare both class fractions

> Compare both class fractions. Apply the supplied Accuracy and Imbalance facts and keep your reasoning.

For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Explain a changed class mix

> Explain a changed class mix. Apply the supplied Accuracy and Imbalance facts and keep your reasoning.

For the illustrative 8-clear 8-blocked set, use baseline [[8,0],[8,0]] and candidate [[6,2],[0,8]]. Calculate ordinary and **balanced accuracy** for each. Explain what stayed fixed and what this example does not prove.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Assess a new comparison independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Assess a new comparison independently. Apply the supplied Accuracy and Imbalance facts and keep your reasoning.

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Limit claims from missing classes

> Limit claims from missing classes. Apply the supplied Accuracy and Imbalance facts and keep your reasoning.

An evaluation contains ten clear cases and no blocked cases. All **prediction**s are clear. State ordinary **accuracy**, explain why blocked performance is unavailable, and describe how our code handles the missing class.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Accuracy and Imbalance

> Show how a common healthy class can hide complete failure to detect spotted leaves. Explain the labelled step before checking the changed condition.

Show how a common healthy class can hide complete failure to detect spotted leaves.

Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy. A candidate hasTP2, FN 0, FP 3, TN 15.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/16

Candidate decisions

	predicted spotted	predicted healthy
actual spotted	2	0
actual healthy	3	15

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/16

Accuracy and Imbalance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Score trivial baseline

Always healthy is 18/20=90% accurate but detects0/2 spotted leaves.



2. Score candidate

Candidate accuracy is 17/20=85%; spotted recall is 2/2=100%.



3. Compare consequences

Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

CHANGE ONE THING

The candidate's three false alarms are corrected without changing its detections.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Show how a common healthy class can hide complete failure to detect spotted leaves.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

1. Compare consequences

2. Score candidate

3. Score trivial baseline

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare consequences Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. Read the picture in order.

Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The candidate's three false alarms are corrected without changing its detections. Which conclusion is supported by the supplied facts?

1. Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set. False alarms move from FP to TN; neither TP nor FN changes.

Repair a missing building block

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training accuracy** replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

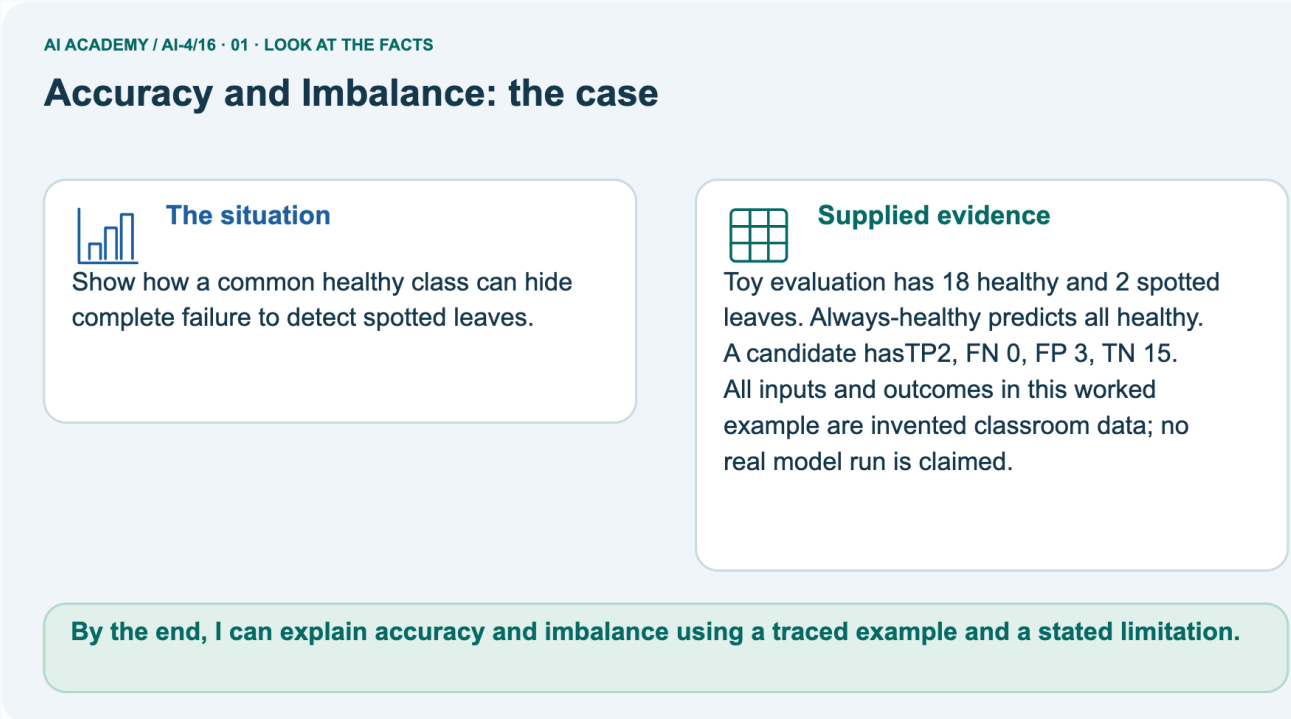
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

Explain each Accuracy and Imbalance picture from its labels, then record a relevant limit.

01. Accuracy and Imbalance: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

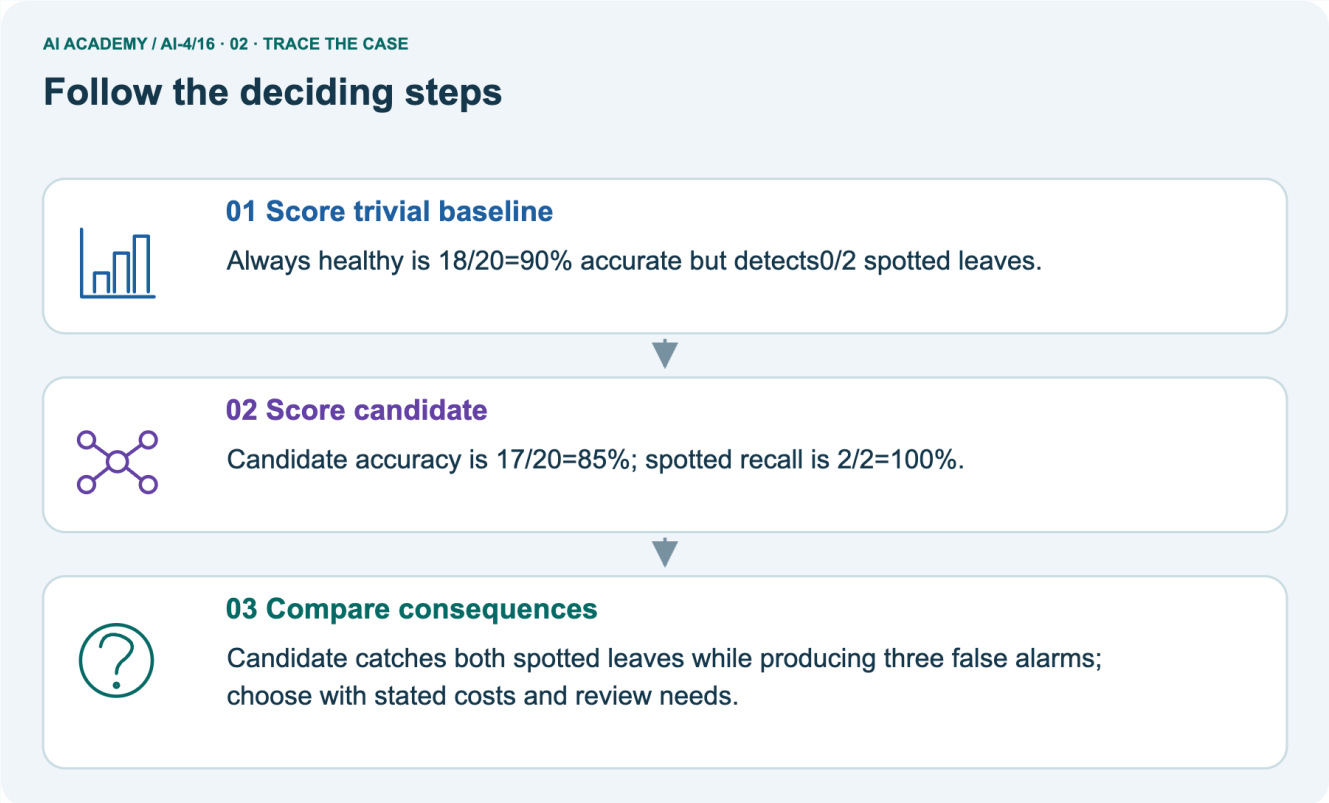
Show how a common healthy class can hide complete failure to detect spotted leaves. Supplied facts: Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy.

A candidate hasTP2, FN 0, FP 3, TN 15. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



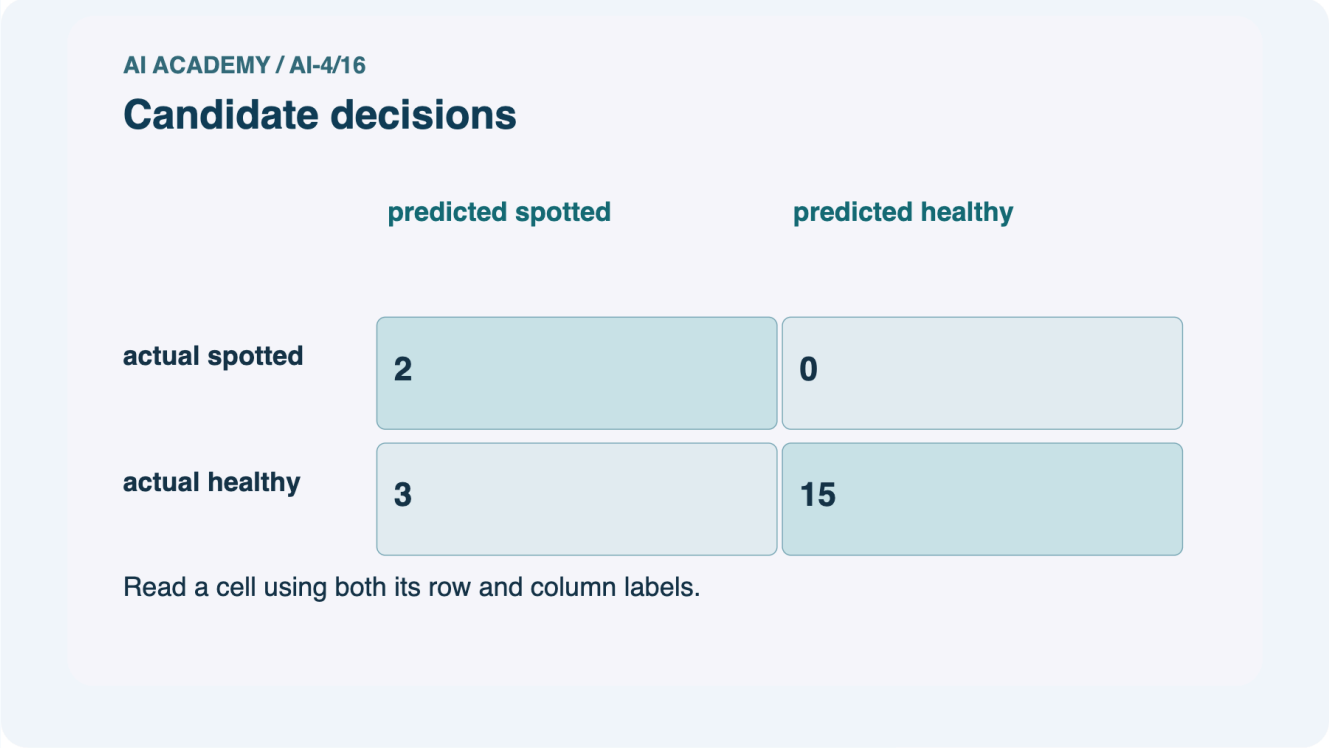
Follow the labelled picture with the matching explanation. Open the vector diagram

Score trivial baseline: Always healthy is 18/20=90% accurate but detects0/2 spotted leaves. → Score candidate: Candidate **accuracy** is 17/20=85%; spotted **recall** is 2/2=100%. → Compare consequences: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Candidate decisions



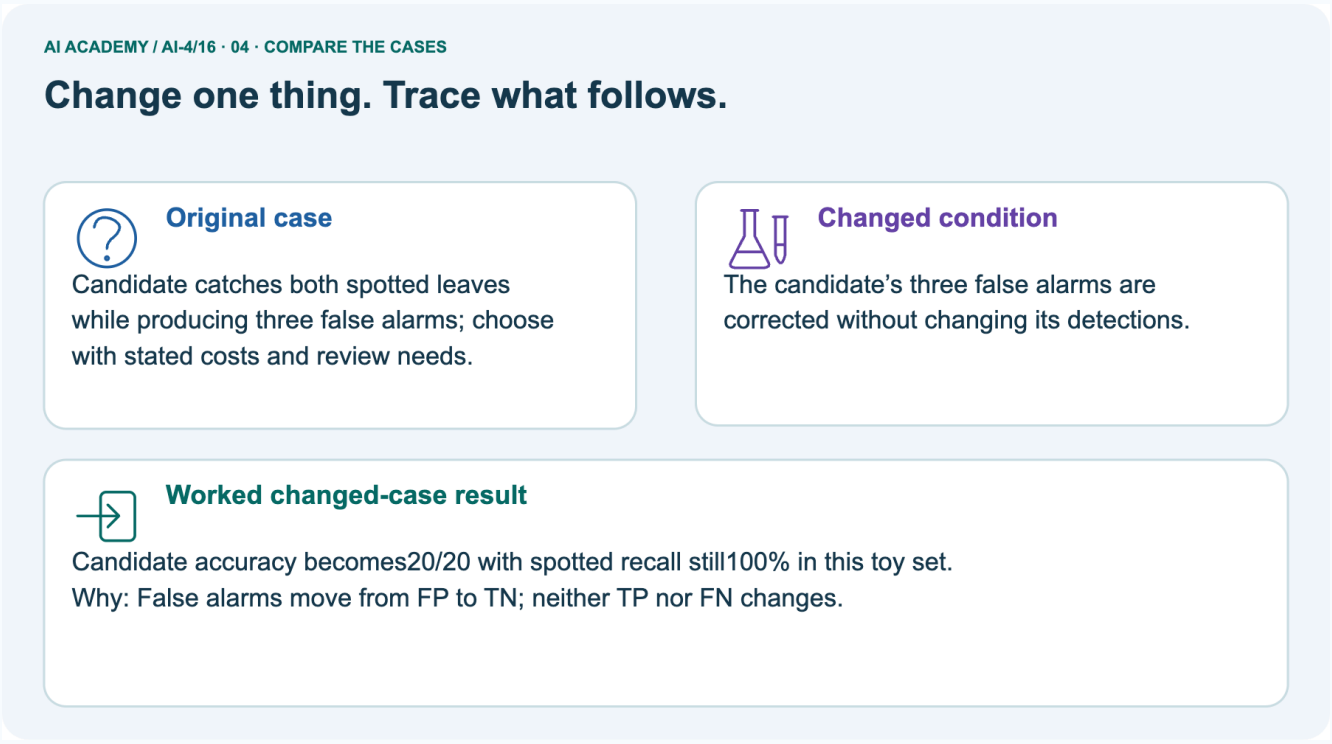
Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate decisions. actual spotted: predicted spotted = 2, predicted healthy = 0. actual healthy: predicted spotted = 3, predicted healthy = 15.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

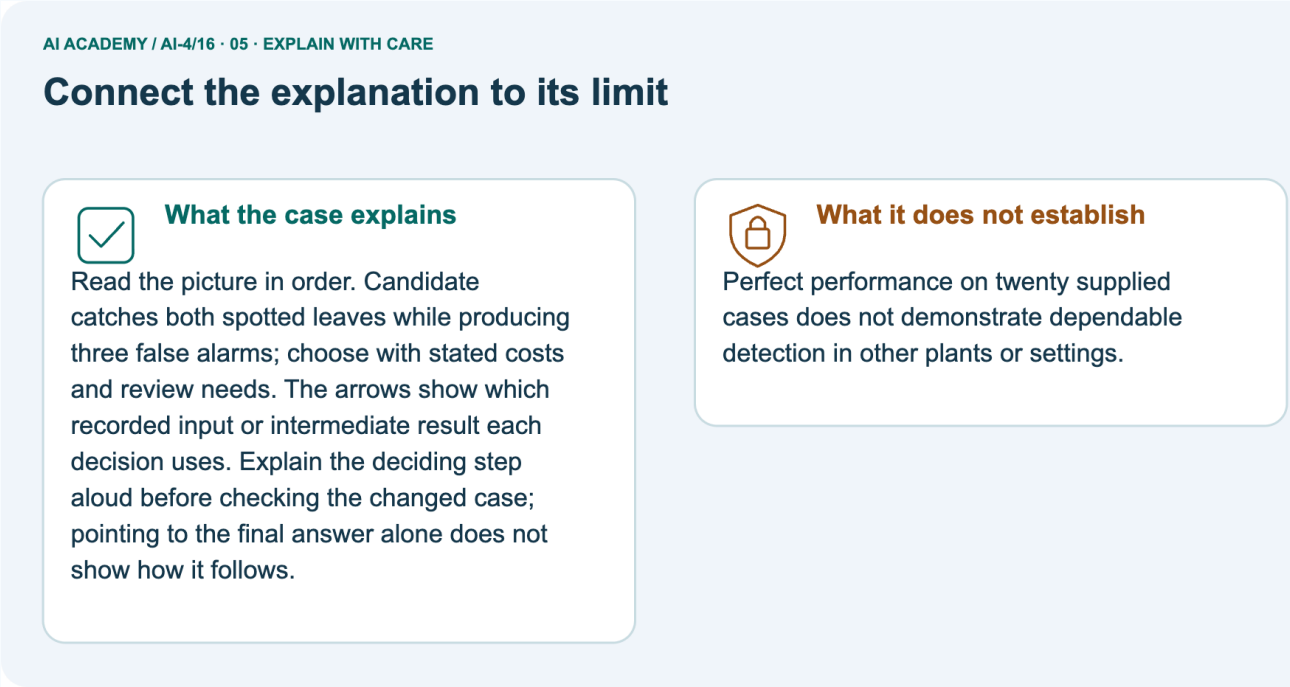
Changed condition: The candidate’s three false alarms are corrected without changing its detections. Expected result: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

Reason: False alarms move from FP to TN; neither TP nor FN changes.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

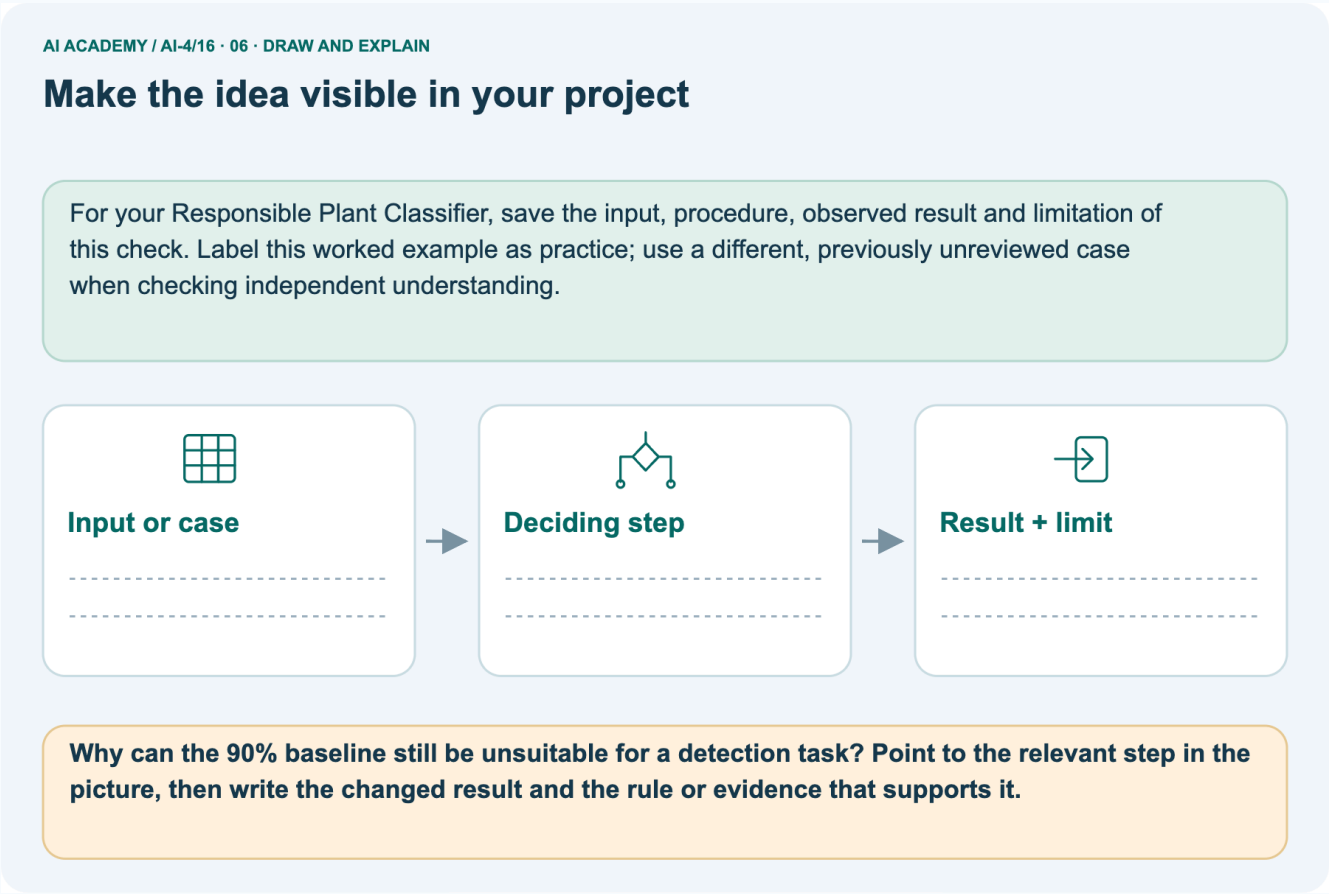
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.